

Question 15.5 of arXiv:1011.1524: the metrizable case is answered negatively

Literature-status note for arXiv:1011.1524

13 August 2026

Status. Literature already answered (metrizable case). Domínguez and Tarieladze explicitly cite Question 15.5 and prove that no metrizable group can have the requested property. Their paper leaves the nonmetrizable version open. This note records a literature identification, not a new proof.

1 The source question

Dikranjan, Shakhmatov, and Spěvák ask the following on PDF page 31, Question 15.5, of arXiv:1011.1524v1 [1]:

Does there exist a (separable metric) group which contains an f_ω -productive sequence but does not contain any unconditionally f_ω -Cauchy productive sequence?

Here f_ω is the constant weight taking value ω ; thus all integer multipliers are allowed. In the terminology of the supporting paper, f_ω -productive means hyper-multipliable, and unconditionally f_ω -Cauchy productive means unconditionally hyper-premultipliable.

2 The explicit literature answer

Domínguez and Tarieladze's later paper *Metrizable TAP, HTAP and STAP groups* [2] identifies the source question by paper and number. On PDF page 11, Remark 2.16, it reproduces Question 15.5 and states that Corollary 2.15(b) gives a negative answer in the metrizable case.

The mechanism appears immediately above the remark. Proposition 2.14 proves that every hyper-convergent sequence in a metrizable topological group has an unconditionally hyper-premultipliable subsequence. Since every hyper-multipliable sequence is hyper-convergent, Corollary 2.15(b) concludes:

If a metrizable group contains an injective hyper-multipliable sequence, then it contains an injective unconditionally hyper-premultipliable sequence.

Translating the terminology back gives

No metrizable—hence no separable metrizable—group satisfies Question 15.5.

The identification is explicit rather than inferred: Remark 2.16 names Question 15.5, gives both terminology correspondences, and states the negative answer.

3 Scope and search status

The parenthetical wording in the source also permits a broader reading asking about arbitrary Hausdorff topological groups. Domínguez and Tarieladze say in Remark 2.16 that they do not know whether Corollary 2.15(b) remains true for nonmetrizable groups. Therefore the precise disposition is

metrizable case: answered negatively; nonmetrizable case: not settled by the cited theorem.

A bounded search through 13 August 2026 checked the local arXiv corpus, exact question phrases, the source’s citation trail, and later papers on productive, hyper-multipliable, TAP, HTAP, and STAP sequences. It located the direct 2012 answer above but no primary-source claim resolving the nonmetrizable variant.

References

- [1] D. Dikranjan, D. Shakhmatov, and J. Spěvák, *Productivity of sequences with respect to a given weight function*, Topology Appl. 158 (2011), 298–324; arXiv:1011.1524.
- [2] X. Domínguez and V. Tarieladze, *Metrisable TAP, HTAP and STAP groups*, Topology Appl. 159 (2012), no. 9, 2338–2352, doi:10.1016/j.topol.2011.09.046.